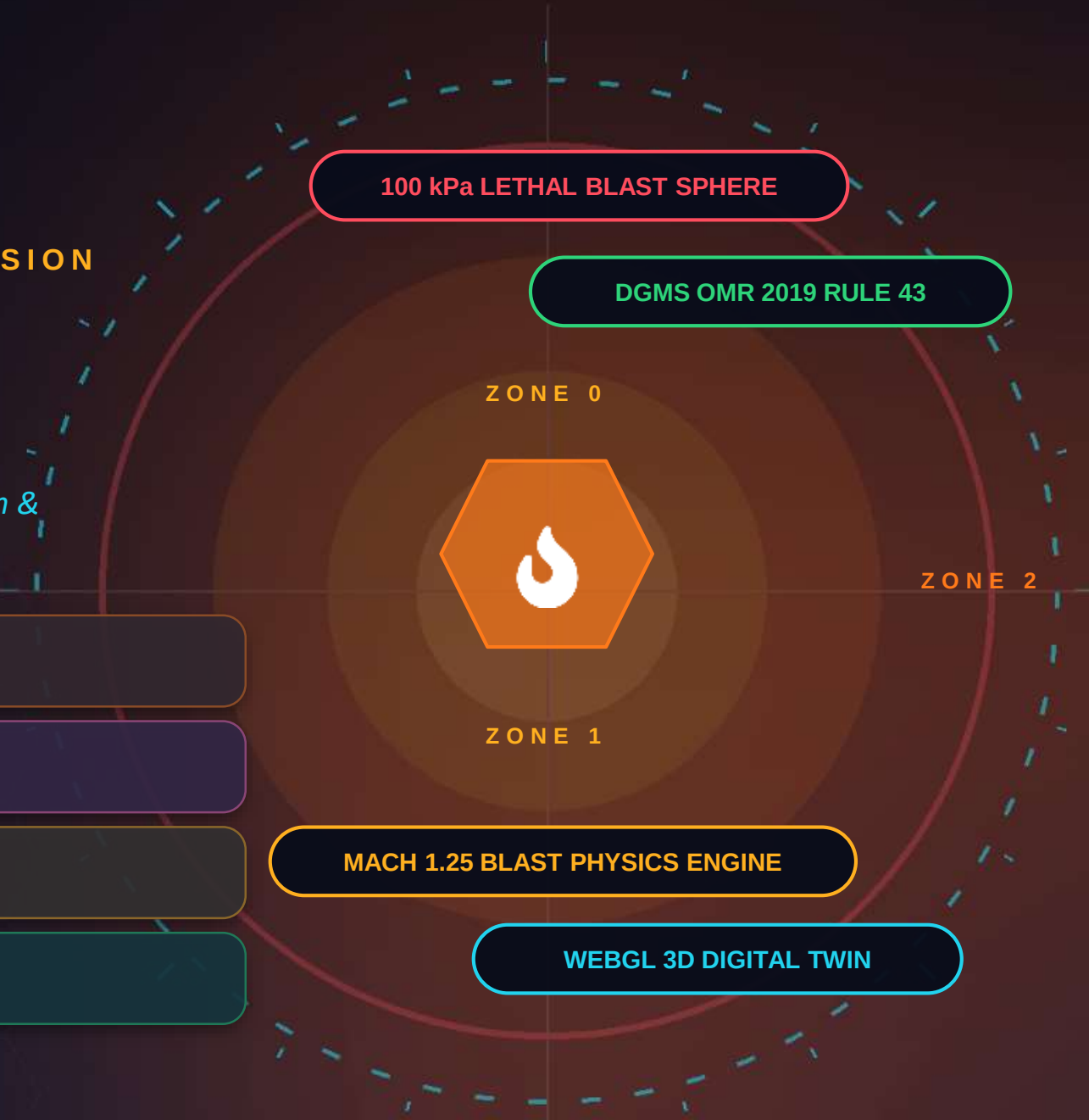
SMART INDIA HACKATHON 2026

SMART INDIA HACKATHON 2026 • IDEA SUBMISSION

SentinelSIF

Bridging the SIF Paradox: Autonomous Hydrocarbon Precursor Detection & Thermodynamic Blast Defense



PROBLEM STATEMENT ID
SIH26165

PS CATEGORY
Software (IoT & SCADA)

TEAM ID
162044

TEAM NAME
Visual Vaults

PROBLEM STATEMENT TITLE
AI-Powered SIF Precursor Prevention & Decision Intelligence Platform

THEME
Smart Automation / Disaster Management / Clean & Green Energy

VISUAL VAULTS

TEAM ID 162044

IDEA TITLE & SOLUTION

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THE PROBLEM: THE SIF PARADOX

80% of fatalities stem from high-energy precursors, not minor slips

72 h near-miss paperwork lag while micro-leaks climb toward LEL

0 physics context: a stair trip equals a pump seal bypass

OUR SOLUTION: SENTINELSIF ENGINE

5-Stage Multimodal Swarm

Parses audio, field text & thermal scans in under 1.2 s to extract chemicals and barrier failures.

Thermodynamic Blast Scaling

Real-time Hopkinson-Cranz overpressure, TNT equivalence & lethal blast radii.

Interactive 3D Digital Twin

WebGL twin shows 100 kPa mortality spheres and Zone 0/1/2 boundaries.

DGMS Rule 43 Action Gateway

Instant, binding Stop-Work orders sent to the Concerned Safety Official.

INNOVATION & UNIQUENESS

Not a passive reporting portal: SentinelSIF fuses Generative AI with first-principles chemical physics to compute real destructive potential, then triggers statutory action before containment is breached.

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TECHNICAL APPROACH

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INGESTION

- Voice memos, field text & operator forms
- SCADA / Modbus, OPC-UA telemetry
- FLIR thermal & camera feeds

AGENT SWARM

- Extractor: CH_4, H_2S, LPG, P&ID nodes
- Physicist: energy & blast overpressure
- Auditor: LOPA / Bow-Tie barriers
- Statutory: DGMS Rule 43 check

3D DIGITAL TWIN

- Three.js / WebGL spatial engine
- 100 kPa lethal blast sphere
- Zone 0/1/2 ATEX hazard heatmap
- Pump P-102, Horton Sphere nodes

ACTION & AUDIT

- 1-click Stop-Work notice
- Dispatch to Concerned Safety Official
- Tamper-proof cryptographic audit trail
- Statutory PDF export in 1 click

CORE TECHNOLOGY STACK

Next.js 16 • React 19 | Three.js / WebGL | Python FastAPI | LangGraph Swarm | PyTorch | NumPy Physics | TailwindCSS

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FEASIBILITY AND VIABILITY

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ENGINEERING FEASIBILITY

< 1.2 s

Verified prototype: live twin + multi-agent analysis

100%

Offline on offshore IPCs (Ollama / Llama 3)

Zero-friction integration

Adapters for Honeywell Experion, Yokogawa CENTUM, SAP EHS & IBM Maximo.

Statutory alignment

Pre-calibrated to DGMS OMR 2019 and OISD standards: no regulatory friction.

POTENTIAL CHALLENGES → MITIGATIONS

01 Noisy / incomplete field text

FIX Multimodal fallback still extracts chemicals, pressures & equipment tags from a one-line report.

02 Operator alert fatigue

FIX SIF-Precursor Energy Gate mutes trivial events; alarms fire only above safe blast-energy limits.

03 Marine environment & link drops

FIX Edge-first Docker deployment with async queuing; auto-syncs when connectivity returns.

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IMPACT AND BENEFITS

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SOCIAL & WORKER

150,000+

frontline crew protected

- Multilingual voice reports: Hindi, Assamese, English
- Catches latent energy before workers enter explosive zones

ECONOMIC & OPERATIONAL

95%

faster triage: 72 h → 90 s

- ₹50+ Cr averted per prevented blowout or VCE
- 100% auditable Rule 43 compliance: zero penalties

ENVIRONMENTAL & ESG

CH_4

fugitive venting stopped at source

- Prevents offshore crude slicks & coastal damage
- Strengthens OIL's Scope 1 ESG & carbon posture

PRIMARY BENEFICIARIES

- Frontline offshore technicians & drillers
- Statutory safety officers & rig managers
- Public-sector energy majors: OIL, ONGC, GAIL, IOCL

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RESEARCH AND REFERENCES

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STATUTORY REGULATIONS & STANDARDS

- DGMS OMR 2019: Rule 43, Stop-Work Authority
- OISD-GDN-169 & OISD-STD-105: hazard ID, risk assessment, work permits
- ISO 45001:2018: OH&S management systems

INDUSTRIAL SAFETY RESEARCH

- Campbell Institute / NSC: The SIF Paradox (2018)
- CCPS (AIChE): LOPA & Bow-Tie models
- Martin & Black (2015): Preventing Serious Injuries and Fatalities

BLAST THERMODYNAMICS & SCALING

- Hopkinson-Cranz law: Z = R / E^(1/3) for vapor explosions
- Baker-Strehlow-Tang model: VCE flame speed & shock progression

HISTORIC DISASTER INQUIRIES

- Piper Alpha (1988): Lord Cullen Inquiry
- Deepwater Horizon (2010): National Commission Report
- Baghjan Well 5 Blowout (2020): OIL technical inquiry